

— PRIVATE DATA · PRIVATE VERIFIER · ON-PREMISES

Evaluating **Qwen3.8** agentic search on a private verifier

No data leaves the premises. The corpus, the verifier, the solver and the judge all run on hardware you already own.

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THIS DECK



A public benchmark stops measuring search once the corpus is public.

1 The answer may already be in the weights.

A model that recites a HotpotQA answer without retrieving anything scores the same as a model that searched well. The score no longer separates recall from search.

2 The scaffold is what is actually being scored.

An agentic system is a model plus tools plus a loop. Once a public set is saturated, the reported gain belongs to the scaffold, and nothing in the score says which part moved.

3 The corpus you care about is not in any benchmark.

The retrieval you deploy runs over your documents, in your vocabulary, with your entity names. No public set contains them.

If you want to know whether a model searches **your** corpus well, the benchmark has to be built on **your** corpus.

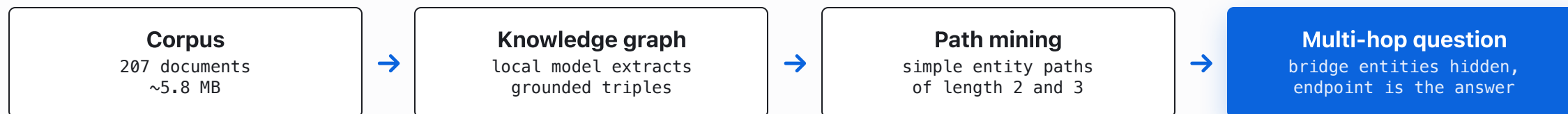
Private data closes the API route for every stage, not just the solver.

CONSTRUCTION	reading every document to mine facts sends the whole corpus through the model.
AUDITING	checking whether an evidence span states its claim sends the span and the source.
SOLVING	the agent under test reads passages, so its context is the corpus.
JUDGING	equivalence grading sees the gold answer and the model answer.

Any one of these on a hosted endpoint puts the corpus outside the premises. So the whole loop is local, and local means a budget GPU rather than a cluster: the study behind this deck ran on single NVIDIA L4 cards with no API spend.

The evaluation and the thing evaluated share one constraint: **everything runs on hardware you own.**

A verifier mined out of the corpus itself.



A path of length k is a chain of k corpus-grounded facts, so path length is the difficulty knob. The question names the starting entity and hides the bridges, so a solver has to discover them rather than look one thing up.

207 documents in the corpus, one organisation's engineering blog	10 graphs built over that same corpus by the same local model	18,615 edges audited, every edge of every graph, no sampling	40 questions sampled per graph for the solver ladder
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Nobody vouches for a benchmark nobody wrote.

When a pipeline emits an evaluation set, acceptance rests on checks the pipeline can compute over its own artifact. The same three recur across systems:

COVERAGE

every source document contributes at least one item.

cheap, exact, model-free

DEDUPLICATION

no repeated items or edges in the artifact.

cheap, exact, model-free

GROUNDNESS

every claim traceable to a span of the corpus, in practice an exact substring check.

cheap, exact, model-free

A run that satisfies all three is reported as having produced a usable evaluation set. The one line of work that inspects acceptance closely, DoRA, abandons automation and sends items to three rounds of domain-expert review.

Before trusting a verifier to rank models, ask what licenses the claim that [the verifier itself is any good](#).

One word, two properties.

PROVENANCE

Is the evidence span an exact substring of the document it cites?
Checked with no model in the loop. This is what automatic pipelines mean by grounded.

a one-word span is trivially a substring

ASSERTION

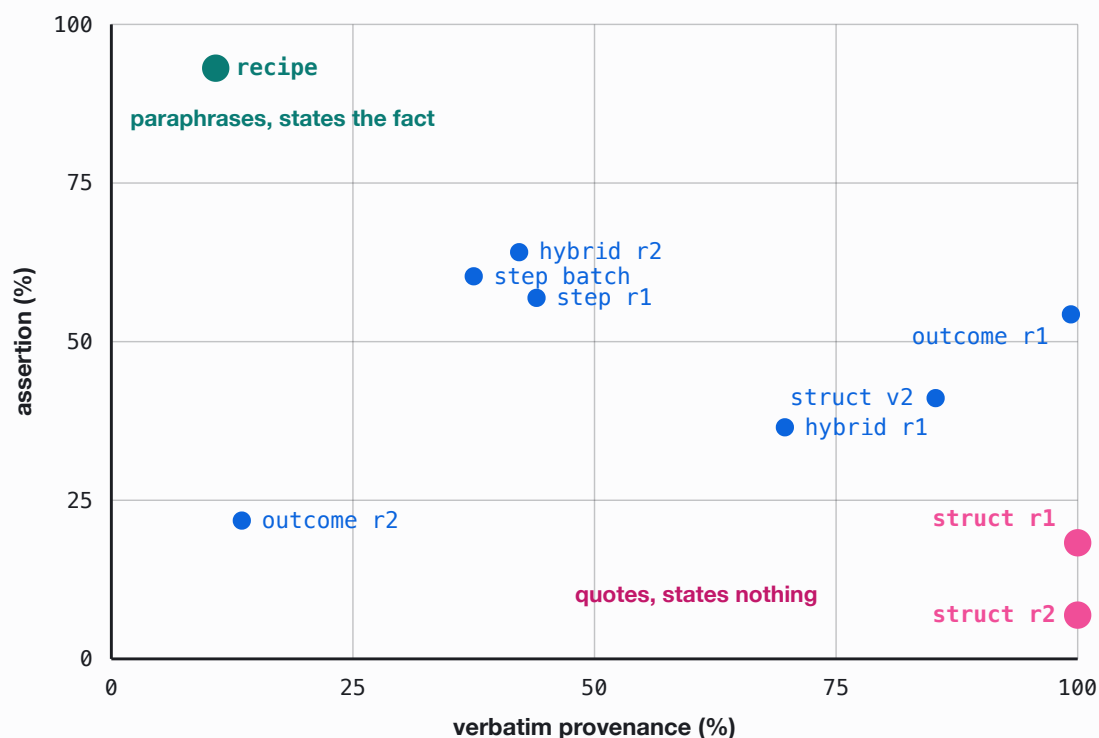
Shown the span alone, can a judge recover the subject, the relation and the object? The subject may be implicit; the relation and the object must be stated.

a keyword without the relation scores negative

Both are cheap. The substring check needs no model. The assertion audit needs one call of at most eight output tokens per edge, 2,811 calls for the largest graph, and all 18,615 edges of all ten graphs were audited at temperature 0 with no sampling.

Standard practice measures the first and reports it as the second.

The two axes invert at the extremes.



Struct r2 is **100% verbatim** and **6.9% asserting**.

median evidence span: 11 characters

Its builder set each evidence span to the object string itself.
Every edge passes the substring check while asserting nothing.

Recipe is **10.8% verbatim** and **93.1% asserting**.

last on provenance, first on assertion

Its spans paraphrase rather than quote, while stating the facts they support. Accepting on substring groundedness keeps the first graph and rejects this one.

A question is worth having when memory cannot answer it and the corpus can.

$$\text{usable}(q) = \neg \text{closed}(q) \wedge (\text{rag}(q) \vee \text{agentic}(q))$$

1

Not answerable closed-book.

Excludes items answerable from parametric memory, which measure recall rather than search.

2

Answerable from the corpus.

Excludes items that are broken, unanswerable, or whose gold answer is simply wrong.

Neither failure mode is visible to any artifact-level check. A graph's **usability** is the fraction of its sampled questions that are usable, and it is the quantity every acceptance metric has to be judged against.

Three conditions over one corpus.

CLOSED-BOOK	No retrieval at all. The model answers from what it already knows.	must fail
SINGLE-SHOT RETRIEVAL	BM25 top-5 passages placed in context, one shot, no follow-up query.	either may succeed
AGENTIC RETRIEVAL	The model issues its own BM25 queries, up to four rounds, before it answers.	either may succeed

Grading is string containment, falling back to an LLM equivalence judge. The ladder is applied identically to every graph and every question, with no graph-specific tuning, because it is the measuring instrument rather than a result.

The closed-book rung is the one that makes the set measure **search** instead of memory.

The most widely used check ranks the artifacts close to inversely.

ACCEPTANCE METRIC	PEARSON R	SPEARMAN P
Document coverage	−0.34	−0.42
Duplicate-free	−0.18	−0.22
Groundedness, verbatim	−0.74	−0.74
Edge count	+0.04	0.00
Groundedness, entailment	+0.05	+0.25
Capacity, log10 paths	+0.29	+0.15
Entity-entity share	−0.01	+0.01

Coverage, duplicate-freedom and edge count carry no usable information. Verbatim groundedness is the strongest correlate in the sample, with a negative sign: the three highest-provenance graphs are the three least usable.

The proposed substitutes are better behaved in sign and too weak to replace measurement. Capacity is dependable only at zero, where the one graph with no length-3 path yielded **one usable question in forty**.

n = 10 graphs. Ten points support ordinal statements, not precise coefficients.

A question set whose difficulty is a property of the corpus.

UNCONTAMINATED

Every item survived a closed-book probe on the same model family, so no item is answerable from parametric memory.

measures search, not recall

ANSWERABLE

Every item was reached by at least one retrieval-equipped solver, so a failure is the solver's rather than the question's.

a conservative lower bound

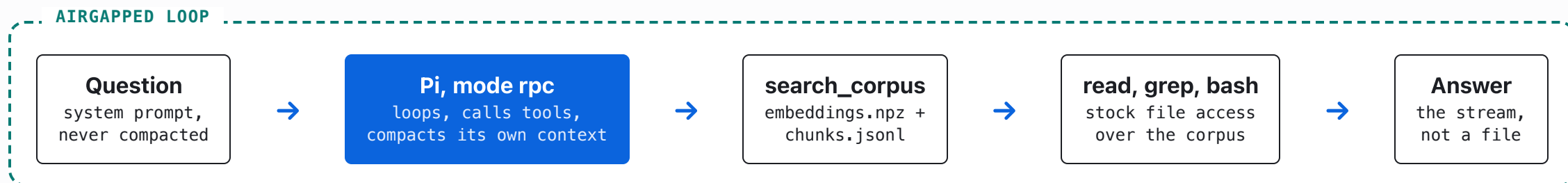
TUNABLE

Path length is the difficulty knob, and it is defined on graph structure rather than on question phrasing.

two hops and three hops

The same protocol reconstructs on any corpus. That is what makes it a verifier rather than a dataset: it is the procedure, applied to documents that never leave the building, and the 207-document corpus here is a shareable stand-in for the private one.

One agent, one external tool, no scaffold to credit.



WHAT THE HARNESS ADDS

A per-run binding of Pi to an OpenAI-compatible /v1, one extension registering `search_corpus`, and two wall-clock rails. No skills, no tool catalog, no supervision of what the model does with its turn.

WHAT IS SWITCHED OFF

`search_web` and `read_url` exist and are off by default, per run. With web access off the extension is never loaded and the model never sees the two tools.

A hit is a starting point, not an answer.

```
// one hit from search_corpus
{ score, path, lines, text }

// what the agent is told to do with it
read path around lines
widen the window while truncated
grep -n the same file for the topic
```

7777 x 768 matrix loaded in process: 117 ms to load, 4.3 ms per search.

The retrieval being tested is exactly one script, and everything else the agent does is stock file access. Nothing sits in between to credit or blame.

The instruction is load-bearing. The `lines` field alone was not enough: runs answered from the snippet without ever opening the file.

The extension is checked against the reference CLI on 15 query and filter combinations, including CJK: identical rankings, maximum score delta 1e-6.



— THE MODEL UNDER TEST

Qwen3.8-27B is practical on a 24 GB card because three layers in four hold no KV cache.

- | ARCHITECTURE 64 layers, `full_attention_interval` 4. Sixteen layers maintain a KV cache; the remaining 48 use linear attention.
- | DRAFT HEAD a native MTP head, `mtp_num_hidden_layers` 1, used for self-speculative decoding.
- | QUANTIZATION Unsloth Dynamic v3.0 UD-Q4_K_XL, 17.92 GB of weights resident.
- | POSITIONS the model supports up to 262,144; the card is what decides where it actually stops.

Context costs roughly a quarter of what it costs on a conventional dense model of this size, which is the whole reason a 27B-class model is being considered for an agentic loop on one budget GPU.

The context ceiling is 104,192 tokens, found by binary search and confirmed under load.

--CTX-SIZE	VRAM USED	FREE
65,536	21,500 MiB	3,070 MiB
90,112	23,138 MiB	1,432 MiB
98,304	23,684 MiB	886 MiB
102,400	23,956 MiB	614 MiB
104,192	24,076 MiB	494 MiB
104,448	fails to serve	

24,570 MiB device. Weights are 17.9 GB at every setting.

A context counts as working only when the server both reaches /health and completes a generation. llama.cpp allocates the KV cache at load time while some compute buffers are sized on the first decode, so a context that loads can still fail under traffic.

A 101,815-token prompt filling 98% of the window returned a correct answer at 24.88 tok/s decode and 278 tok/s prefill.

VRAM peaked at 24,082 MiB, container stayed healthy

Binary search to a tolerance of 256 tokens. ECC off returns about 1 GB, and the configuration does not fit without it.

Twenty-three tokens a second, and the arithmetic says why.

WORKLOAD	DECODE TOK/S	MTP ACCEPT.
summarization	24.63	0.810
math	24.54	0.803
multi-turn	23.69	0.798
code	23.67	0.810
chat	23.24	0.762
prose	22.96	0.736
json	22.92	0.725

The L4 provides 300 GB/s. Reading 17.9 GB of weights once per token bounds plain autoregressive decode at about **16.8 tok/s**. Everything above that line is the MTP draft head.

Speculative decoding is lossless by construction, and 40 arithmetic problems with known ground truth score 40/40 with and without it.

The sparse alternative on the same card is much faster: Qwen3.6-35B-A3B sustains **92 to 100 tok/s** at a 56,320-token context, because it activates roughly 3B parameters per token.

Greedy, cache disabled, 256 output tokens, decode only, averaged over two runs.

An agentic turn is mostly prefill, and prefill is what the window pays for.

104,192 token context on the model under test, the measured card ceiling	56,320 token context on the 35B-A3B alternative, on the same class of card	23 tok/s decode, versus 92 to 100 for the sparse alternative	278 tok/s prefill at 102K, where cost amortises over a long batch
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An agentic search loop spends its budget re-reading: each round appends tool output and re-prefills a growing prefix. The window bounds how many rounds fit before compaction starts discarding evidence, and decode speed bounds how long the whole thing takes. The two models sit on opposite sides of that trade, which is exactly why the comparison is worth running rather than reasoning about.

Whether the longer window or the faster decode wins on this verifier
is **an empirical question**, and it is the one this experiment asks.

Two axes, held fixed everywhere else.

		CLOSED-BOOK	SINGLE-SHOT BM25	AGENTIC BM25, 4 ROUNDS
Qwen3.8-27B	104,192 ctx	control: excludes items answerable from memory	retrieval without a loop	the condition under test
Qwen3.6-35B-A3B	56,320 ctx	control, same corpus	retrieval without a loop	the incumbent to beat

HELD FIXED

The corpus, the question set, the mining and verbalization stage, the retrieval index, the grading procedure, and the four-round cap. Only the served model changes across rows.

SAMPLE

Per-graph samples are 40 questions, fewer where a graph is too sparse to fill the sample, so per-graph rates carry sampling error of several points.

Accuracy per turn, against two different budgets.

ANSWER SNAPSHOT	The answer the model holds at the end of every turn is captured, and carried forward on turns where it did not change.	per turn
JUDGE	Every turn's answer is graded against ground truth by the same local judge, not only the final one.	per turn
X-AXIS, TURNS	Accuracy against turn index. The legible unit: one turn is one round of the loop.	1 to budget
X-AXIS, TOKENS	Accuracy against cumulative fresh-prefill input tokens, summed compaction-safe from the session log.	the real budget

Turns are what a person reads; fresh-prefill input tokens are what the GPU pays. A model with a longer window can spend more tokens per turn, so the two axes can disagree about which model is more efficient, and both are reported.

The table exists. The cells do not.

MODEL, AGENTIC CONDITION	CLOSED-BOOK	SINGLE-SHOT	AGENTIC	TURNS TO ANSWER	FRESH-PREFILL TOKENS
Qwen3.8-27B-UD-Q4KXL-MTP					
Qwen3.6-35B-A3B-Q4KXL-MTP					
Difference	not yet measured				

 hatched cell = designed, not yet measured

Nothing on this machine contains a Qwen3.8 accuracy on this verifier. The rows, the columns and the metric definitions are fixed in advance so that the numbers cannot be chosen after the fact; the cells stay empty until the run happens.

● AWAITING MEASUREMENT

Three outcomes, three different conclusions.

QWEN3.8 AHEAD ON ACCURACY

The longer window is worth four times the decode cost on multi-hop retrieval, because more evidence survives to the answering turn.

then: window > throughput for this loop

35B-A3B AHEAD AT EQUAL WALL TIME

More turns beat a bigger window: the loop recovers by querying again, and compaction costs less than it appears to.

then: throughput > window for this loop

BOTH FLAT ABOVE TURN TWO

The verifier is not exercising the loop. Either the questions resolve in one retrieval or the four-round cap binds before the model needs it.

then: the instrument, not the models

The third outcome is the one worth stating out loud in advance, because it is a result about the verifier rather than about either model, and a result that only becomes visible if the per-turn curve is recorded rather than the final answer alone.

The instrument bounds the claim.

- | ONE CORPUS 207 documents from one organisation in one domain. Ordinal claims, not coefficients.
- | BM25 RETRIEVAL the solver ladder retrieves lexically, so a question unreached by any solver may still be answerable by a stronger system. Usability is a conservative lower bound.
- | FOUR ROUNDS the agentic condition is capped, and the cap is part of the measurement rather than a property of the model.
- | SAME-FAMILY JUDGE the judge comes from the same model family as the builders, which risks self-preference. The task is kept narrow and run at temperature 0.

One further constraint is already measured and worth carrying: across all ten construction runs, no builder ever invoked the local embedding or reranking tools available in its loop. Provided capability that is not directed goes unused, which is a warning about how much of an agentic result belongs to the agent.

**A benchmark you built yourself
is only worth what **a measurement
of its purpose** says it is.**

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THE HARNESS

